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# NBA Player Performance Prediction

A Deep Learning Approach of Game Stats

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# Motivation & Introduction

## The Challenge of Volatility

- **Uncertainty Factors:**

NBA player performance is inherently filled with uncertainty, making consistent prediction difficult.

- **Goal:**

We aim to navigate this noise to find stable signals.

## Deep Learning from Game Stats

- **Beyond Surface Data:**

Investigating whether Deep Learning can extract features from "paper data"

- **Visual Context:**

Exploring whether integrating Spatial Structure (Shot Heatmaps) can reveal additional behavioral patterns.

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# Data Pipeline & Engineering

## 01 Data Collection

Sourcing comprehensive **game statistics** and **spatial shot data** from official NBA sources.

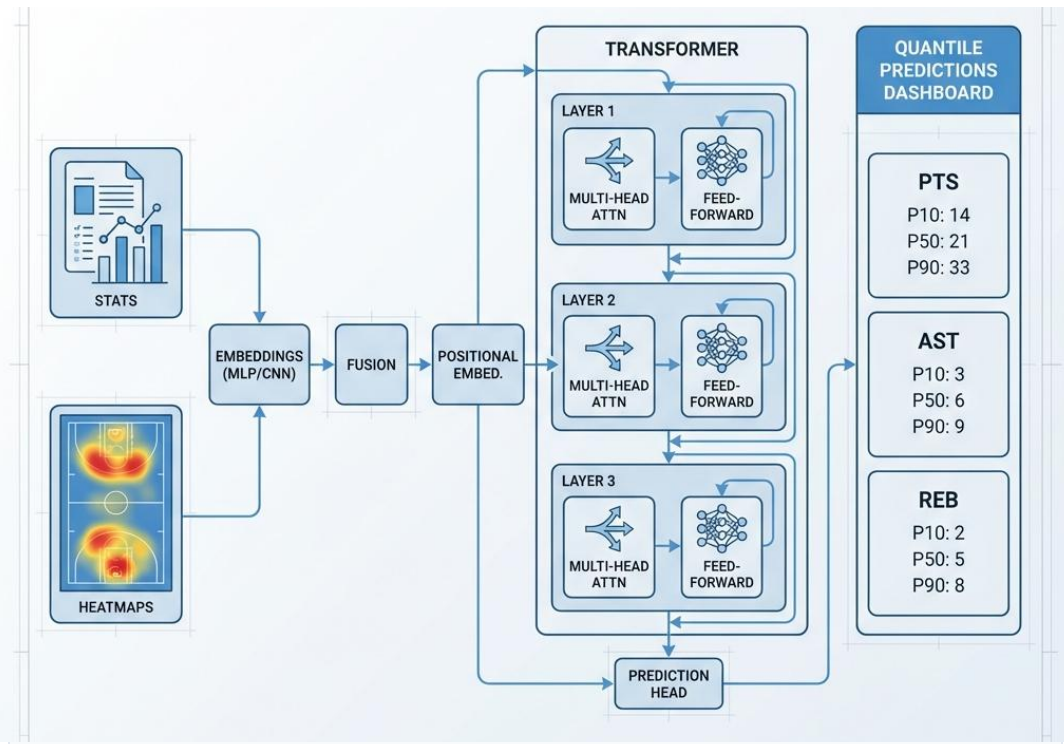
## 02 Feature Engineering

Crafting rolling averages, lag features, and **2D spatial heatmaps** to capture player dynamics.

## 03 Sequence Preparation

Structuring data into **7-game sliding windows** to enable the Transformer to model long-term temporal dependencies.

# Detailed Model Architecture



## Multimodal Encoding

**Visual (CNN):** Extracts spatial features from 2D shot heatmaps to capture shooting tendencies.

**Statistical (MLP):** Creates non-linear embeddings.

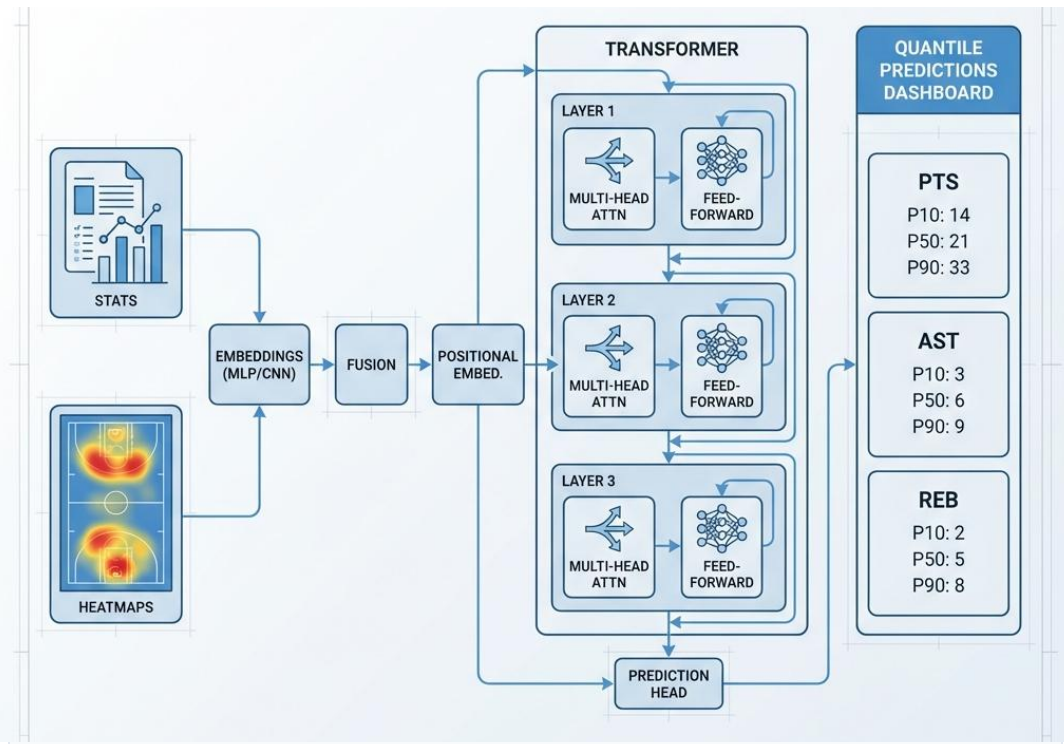
## Transformer Core

**Fusion Layer:** Projects combined embeddings into a unified latent space.

**Positional Embedding:** Injects temporal order information to preserve the sequence of games.

**Attention Layer:** Identifies key correlations and dependencies across the 7-game history.

# Detailed Model Architecture



## Confidence Modeling (Quantile Regression)

### Triple-Output Design:

#### P10 (Floor):

The conservative estimate (worst-case scenario).

#### P50 (Median):

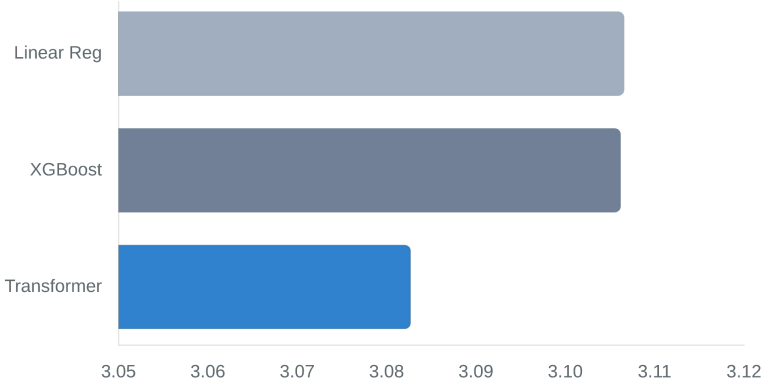
The most likely outcome (expected performance).

#### P90 (Ceiling):

The optimistic estimate (best-case potential).

# Experimental Results 1: Model Comparison(7-dim Input)

| Model Architecture      | MAE (Lower is Better) |
|-------------------------|-----------------------|
| Naive (Historical Mean) | 4.3798                |
| Linear Regression       | 3.1066                |
| XGBoost (Tree-based)    | 3.1062                |
| Multimodal Transformer  | 3.0827                |



## Key Observation

The Transformer achieves the lowest Mean Absolute Error (MAE), shows that it contracts more features from low-dim input.

# Experimental Result 2: Impact of Spatial Heatmaps

| Model Configuration   | MAE (Lower is Better) |
|-----------------------|-----------------------|
| No CNN (Stats only)   | 3.0827                |
| With CNN (Multimodal) | 3.0866                |
| Experimental Result   | No Significant Gain   |

## 1. Information Redundancy

Traditional stats (FG%, 3P%) may already implicitly describe the "where" and "how" of a player's shot profile, making spatial data redundant.

## 2. Relevance Gap

Spatial patterns are crucial for individual shot outcomes but less critical for aggregate game-total counts like PTS or REB.

## 3. Compression Loss

Compressing complex 2D image data to match statistical feature dimensions likely stripped away critical spatial information during fusion.

# Betting Mechanics

## OVER / UNDER

Betting on the **Total Score** of both teams.

- **Over:** Above the line
- **Under:** Below the line

Ex: O/U 10.5 points

## ODDS ORIGIN

Set by bookmakers using:

1. **Probability** Models
2. **Market Balance** (Money flow)

## THE "VIG" (JUICE)

The house's commission hidden in the odds.

|                 |             |
|-----------------|-------------|
| Fair (No Vig)   | 2.00 / 2.00 |
| Real (With Vig) | 1.90 / 1.90 |

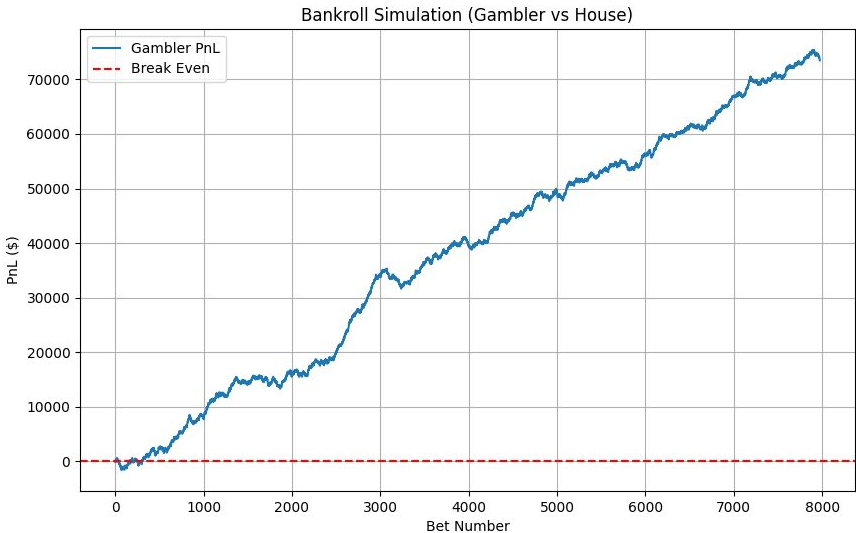


# Experimental Result 3 : Betting Simulation & ROI Analysis

## Betting Simulation Results

| Metric         | Our Model Result |
|----------------|------------------|
| Win Rate       | 54.61%           |
| ROI            | 9.22%            |
| Total Bets     | 7,973            |
| Final Bankroll | \$83,500         |

- Targeting **High-Confidence (Narrow Spread)** games effectively filtered out volatility. This strategy boosted the **win rate** to **54.61%**, proving the model's ability to identify profitable, high-probability edges.



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## Experimental Result 3 : Betting Simulation & ROI Analysis

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| Model / Strategy        | Bets | Win Rate | Realized ROI |
|-------------------------|------|----------|--------------|
| Hybrid (Alpha + Spread) | 69   | 65.2%    | +20.7%       |
| MultiModal Transformer  | 633  | 55.9%    | +4.8%        |
| Naive (Historical Mean) | 770  | 53.7%    | -0.1%        |
| XGBoost                 | 770  | 53.5%    | -0.6%        |
| Linear Regression       | 770  | 51.7%    | -3.8%        |



- The results validate that integrating Quantile Confidence successfully transforms the model from a raw statistical predictor into a Risk Management Tool.

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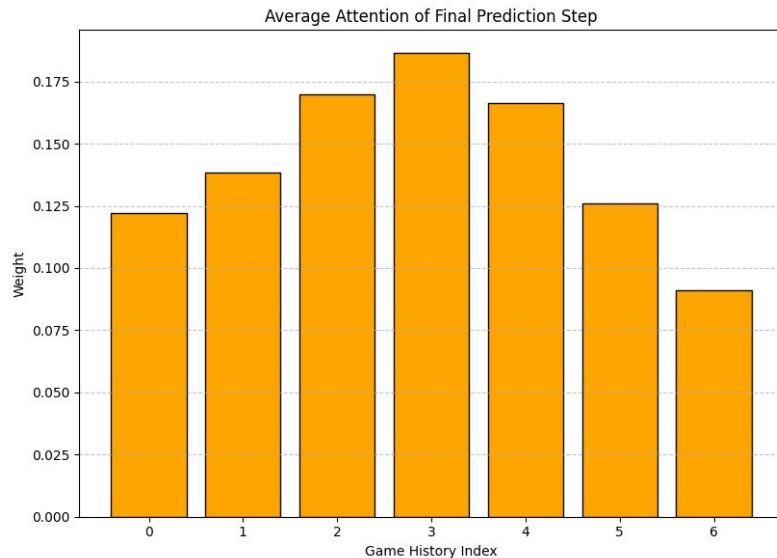
# Attention Map Analysis

## "Bell-Shaped" Attention

Visualizing temporal dynamics reveals that weights are often highest in the middle of the 7-game sequence (games 3-5), rather than the most recent game.

## Stability over Recency

The model relies on a "stabilized average form" to avoid overreacting to extreme outliers in the very last game, identifying critical "Anchor Games" for prediction.



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## Conclusion & Future Directions

### Architecture Efficiency

Deep Learning demonstrates superior automated feature extraction, mining complex patterns from raw data that traditional models require extensive manual engineering to achieve.

### Strategic Success

Quantile regression is the most critical component, transforming a prediction model into a risk-management tool that enables stable long-term ROI in volatile environments.

### Future: NLP Integration

Integrating Large Language Models (LLMs) to analyze play-by-play text and news sentiment will capture contextual nuances like momentum shifts and defensive intensity.